

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO TOOLKIT · CH1 · L12

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The Map

This lesson collapses Ratio Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 5 website questions, 5 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

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WORKED MODEL

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Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 \text{ parts} = 36$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P1H Q4

Ducks pigs

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$9 \text{ parts} = 36$$

Finish: 70

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 On a farm there are chickens, ducks and pigs. The ratio of the number of chickens to the number of ducks is $7 : 2$ The ratio of the number of ducks to the number of pigs is $5 : 9$ There are 36 pigs on the farm. Work out the number of chickens on the farm.

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"work out", "show", "hence", a familiar exam phrase	Use total parts
"work out", "show", "hence", a familiar exam phrase	Convert mixed information to parts
"work out", "show", "hence", a familiar exam phrase	Check the order of the ratio
"work out", "show", "hence", a familiar exam phrase	Split the ratio
"work out", "show", "hence", a familiar exam phrase	Total sailing

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.